



0) Shared Mental Model — The Four Coherence Axes

Purpose

To formalize a **geometric language** for describing, comparing, and optimizing constructive TSP algorithms.

Every greedy heuristic implicitly shapes geometry in **polar coordinates** around some centroid (explicit or emergent).

Thus, four orthogonal coherence axes suffice to characterize its spatial order.

1□ Dualitas — Angular–Radial Balance

Definition

The *equilibrium* between angular progression ($\Delta\theta$) and radial progression (Δr) in the tour sequence.

It measures whether the heuristic alternates directionally and radially in a balanced, harmonious way.

Intuition

- Too angular $\rightarrow$ erratic rotation, “orbit skipping.”
- Too radial $\rightarrow$ long spokes, poor local closure.
- Balanced $\rightarrow$ smooth outward spiraling motion.

Quantification

- Compute ratio $R = \sigma(\Delta\theta)\sigma(\Delta r/r^-)R = \sigma(\Delta r/r^-)\sigma(\Delta\theta)$.
- Coherence increases as variance ratio approaches 1 (balanced energy between radial and angular domains).
- Can also track *local phase alignment* between $\Delta\theta$ and Δr oscillations.

Functional Role

- Core “shape” regulator; sets baseline of geometric stability.
 - Equivalent to *Dualitas* in Blueprint: maintains polarity balance (angle $\leftrightarrow$ radius tension).
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2□ Adaptatio — Radial Smoothness

Definition

Continuity and gradualness of radial expansion — how smoothly the tour evolves outward from the centroid.

Intuition

- Low Δr variance = gentle spiral, stable growth.
- High Δr variance = jagged radial jumps, unstable pattern.

Quantification

- $\text{Adaptatio} = 1 - \sigma(|\Delta r|) \bar{r}$ $\text{Adaptatio} = 1 - \bar{r} \sigma(|\Delta r|)$.
- Optionally weighted by local point density to penalize sudden long hops.
- Can also express as autocorrelation of $r(t)$: high correlation = strong Adaptatio.

Functional Role

- Corresponds to *sustainability of flow*: ensures the spiral behaves like a living structure, not a teleporting pointer.
- Controls local edge smoothness and radial coherence.

3 Textura — Angular Uniformity / Bin Field Quality

Definition

Uniformity and evenness of angular bin occupation — the fabric that holds angular exploration consistent.

Intuition

- In constructive algorithms, we discretize 2π into bins or angular sectors.
- Good Textura → evenly filled bins, consistent angular coverage.
- Poor Textura → clusters of visited points, voids elsewhere (fragmented weave).

Quantification

- Coefficient of variation (CV) of point counts per angular bin:
 $\text{Textura} = 1 - \sigma(b_i) \bar{b_i}$ $\text{Textura} = 1 - \bar{b_i} \sigma(b_i)$.
- Or entropy measure of angular distribution (normalized Shannon entropy).
- For continuous models: variance of angular density over $[0, 2\pi]$.

Functional Role

- Structural scaffolding — how the algorithm maintains angular connectivity and coverage.

- In Blueprint logic, *Textura* = the mesh that sustains flow (like lattice in biological tissue).

4▣ Ecologia — Rotational Rhythm Coherence

Definition

Consistency of rotational increment patterns — the rhythmic stability of $\Delta\theta$ evolution through the tour.

Intuition

- Perfect golden-angle rotation yields maximal dispersion, minimal angular resonance.
- Deviations form rhythmic “beats” — these can improve or degrade global coherence.
- Measures the temporal (stepwise) harmony of rotation.

Quantification

- Compute sequence of $\Delta\theta_k$ differences; measure their **autocorrelation** or **spectral entropy**.
- High coherence = dominant single frequency (stable rotation); low = noisy, multi-frequency.
- Alternate form: correlation between step index k and cumulative angle α_k .

Functional Role

- Governs global dispersion and long-range balance.
- Connects local structure (Dualitas, Adaptatio, Textura) into a global “rhythmic ecology.”
- In Blueprint, it is the *Synchronia* aspect of spatial rhythm — flow of coherence through time.

5▣ Why These Four Form a Minimal Orthogonal Basis

Axis	Domain	Physical Analog	Orthogonality justification
Dualitas	Angle↔Radius ratio	Directional vs radial tension	Defines phase-space orientation
Adaptatio	Radial continuity	Radial flow smoothness	Independent of angular distribution
Textura	Angular field	Structural fabric / bin uniformity	Independent of local Δr patterns
Ecologia	Temporal rhythm	Rotational oscillation	Independent of instantaneous geometry

Together they span **local–global**, **angular–radial**, and **static–dynamic** coherence.
No pair is redundant; all others (like curvature, clustering) can be expressed as combinations.

6□ Derived / Secondary Measures

Once these four are locked, secondary indicators derive naturally:

- **Resonance (R_e)** = geometric mean of the four coherence scores.
 - **Anisotropy Index (A_i)** = difference between angular and radial variances (from Dualitas).
 - **Smoothness index (S_i)** = combination of Adaptatio × Ecologia coherence.
- These support a universal metric layer for benchmarking across algorithms.
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7□ Implementation Pointers

- Evaluate all four metrics **per tour** and **over time** (during construction).
 - Normalize [0,1] for visual radar / polygon plot comparison.
 - The **GAO Dashboard** should always include a “4-axis radar” beside cost/runtime curves.
 - When comparing versions, look for distinct signatures:
 - v1 → high Adaptatio, stable Ecologia, medium Textura.
 - v7 → higher Textura & Dualitas, more rhythmic patterning.
 - v7+ → maximized resonance across all four.
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8□ Narrative Summary

Dualitas keeps the spiral balanced,

Adaptatio keeps it smooth,

Textura weaves its fabric,

Ecologia sustains its rhythm.

Together, they form the **coherence matrix** — the living geometry of constructive optimization.